

Abstract: Airbnb Price Analysis Using Python

This project presents a detailed analysis of Airbnb rental listings to identify the key factors influencing pricing across different locations. Utilizing publicly available Airbnb datasets, we perform comprehensive data cleaning, exploratory data analysis (EDA), and visualizations using Python libraries such as Pandas, Matplotlib, and Seaborn.

The primary objective of this analysis is to understand how elements like neighborhood, room type, number of reviews, availability, and listing features affect the price of Airbnb accommodations. The study focuses on uncovering:

Average and median pricing trends

High-demand locations and their pricing patterns

The relationship between listing characteristics and rental cost

By visualizing these trends and correlations, the project aims to:

Assist travelers in finding budget-friendly stays

Help hosts set competitive and optimized prices

Offer data-driven insights for tourism and hospitality researchers

This project showcases core data analytics skills, including data preprocessing, feature exploration, and storytelling through visual data. It also provides a foundation for building more advanced pricing prediction models or recommendation systems in future studies.